The aim of this section is to develop an understanding of the fundamental principles of heredity, which are illustrated by reference to model organisms and to the inheritance of genetic diseases.

The concept of epigenetics will also be considered in relation to the increase in diseases such as Type 2 diabetes.

Some of the practical techniques used to analyse and manipulate DNA are also considered, as well as the applications of these techniques. The potential uses of DNA and RNA technology in the treatment of disease are considered, as is the profound ethical implications of gene technologies and the role of genetic counsellors.

This section provides opportunities for the use of statistical tests in analysing data.

Learners are expected to apply knowledge, understanding and other skills gained in this section to new situations and/or to solve related problems.

5.1.1 Patterns of inheritance

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	patterns of monogenic (monohybrid) inheritance	To include the correct usage of the terms gene, allele (gene variant), locus, phenotype, genotype, dominant and recessive, heterozygous and homozygous and codominant. HSW8
(b)	gene mutations	To include cystic fibrosis, sickle cell anaemia, phenylketonuria (PKU) and Huntington's disease.
(c)	patterns of inheritance which show codominance and multiple gene variants (alleles)	To include the inheritance of blood groups and HLA antigens in humans.
(d)	patterns of inheritance which show sex linkage and autosomal linkage	To include haemophilia (sex linkage) and ABO blood group and nail patella syndrome (autosomal linkage).

(e)	the use of model organisms to investigate patterns of inheritance	To include *Drosophila melanogaster* **AND** patterns of inheritance to include dihybrid inheritance **AND** the use of the chi-squared (x^2) test to analyse various patterns of inheritance. *M1.4, M1.9* HSW1, HSW2, HSW3, HSW5, HSW6
(f)	chromosome mutations in humans	To include non-disjunction and translocations in the context of Turner's syndrome, Klinefelter's syndrome and Down's syndrome.
(g)	the role of the genetic counsellor and the ethical issues involved in advising families where a genetic disease has been identified.	To include pedigree analysis to predict the probability of genetic disease **AND** the use of genetic testing. *M1.4* HSW2, HSW5, HSW8, HSW9, HSW11, HSW12